Roll No. ..... Total Pages : 3

9129

(2113)

B.Tech. 3rd Semester (Computer Engineering)

COMPUTER ORGANIZATION AND ARCHITECTURE

Paper–BCSE-502/BCSE-503

(From the Session 2019-20 in CSE and Common with Cloud Technology and Information Security and Data Science From the Session 2021-2022 and From the Session 2023-2024 in Full Stack Development and Big Data and Analytics)

Time : Three Hours] [Maximum Marks : 60

Note : All Sections are compulsory. Marks are indicated against each question.

SECTION – A

1. (i) Why we use RTL?

   (ii) Write the purpose of Address register.

   (iii) Define a macro.

   (iv) Write two advantages of microprogramming.

   (v) Explain the utility of control memory.

   (vi) Why cache memory is needed?

   (vii) Write any three data transfer modes.

   (viii) Explain the term page fault.

(ix) What is the disadvantage of Write back policy for cache update?

(x) Differentiate between static and dynamic memory.

(xi) What do you mean by throuput?

(xii) Write any *two* pipeline hazards. (12×1=12)

SECTION–B

2. List any *two* advantages of non-restoring division over restoring division.

3. Differentiate between privileged and non-privileged instructions.

4. Write two advantages of RISC architectures.

5. Define the term " locality of reference " . List its types. (4×2=8)

SECTION — C

6. Draw and explain the instruction cycle of a computer.

7. With the help of suitable diagram explain the operation of DMA.

8. What are the different challenges faced by superscalar processors?

9. Define associative memory. Derive match logic for one word of associative memory. (4×4=16)

SECTION–D

10. What is the purpose of fast addition? With the help of suitable diagrams explain the operation of Carry Look Ahead adder.

11. What are the advantages offered by hardwired control unit over microprogrammed control unit ? With help of neat diagram explain the operation of micro-program sequencer for control memory.

12. Write any *six* design challenges faced by parallel computer designers. Write and explain the Flynn 's classification of parallel processor systems.

13. What do you mean by hit ratio? Explain any *three* methods to improve the hit ratio. (4×6=24)

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(From the Session 2019-20 in CSE & Common with Cloud Technology & Information Security and Data Science from the Session 2021-2022 and from the Session 2023-2024 in Full Stack Development and Big Data and Anlaytics common with AI & Machine Learning From the Session 2024-2025)

Time : Three Hours] [Maximum Marks : 60

Note : All Sections are compulsory. Marks are indicated against each Section.

SECTION–A

1. (i) Data dependence is bad for pipeline. Comment.

   (ii) Write the purpose of program counter.

   (iii) Define a micro instruction.

   (iv) Write *two* disadvantages of microprogramming.

   (v) Explain the utility of control memory.

   (vi) Define locality of reference.

   (vii) Write any *three* data transfer modes.

(viii) List any *two* advantages of write through cache writing policy.

(ix) What do you mean by addressing mode ?

(x) Write any *one* advantage of CISC processors.

(xi) What do you mean by memory mapped I/O?

(xii) What are different essential components of an instruction? (12×1=12)

SECTION–B

2. What is the purpose of guard bits which are used in floating-point operations?

3. Write any *two* advantages of a stack based CPU design method.

4. Why is "branch taken" not suitable for static prediction?

5. What do you mean by memory interleaving ? Write its *two* advantages. (4×2=8)

SECTION–C

6. Explain why is the two's complement format the most popular one for representing binary numbers.

7. Differentiate between interrupts and exceptions. Also explain the advantages offered by software.

8. What do you mean by a coherent memory system? Discuss any *one* scheme for enforcing coherency.

9. Define the term fragmentation. Compare and contrast internal and external fragmentation. Explain with the help of example, how external fragmentation can be solved with paging. (4×4=16)

SECTION–D

10. Write and explain the Booth's multiplication algorithm. Also illustrate with suitable example.

11. Define a micro-program sequencer. With the help of neat diagram explain the operation of micro-program sequencer for control memory.

12. A computer has only one delay slot. The Instruction in this slot is always executed, but only on a speculative basis. If a branch does not take place, the results of that instruction are discarded. Suggest a way to implement program loops efficiently on this computer.

13. Consider the following page reference string :
1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6
How many page faults would occur for following replacement algorithms assuming 3, 4, 5 frames. All Initially empty.
(a) LRU Replacement.
(b) FIFO replacement.
(c) Optimal Replacement. (4×6=24)